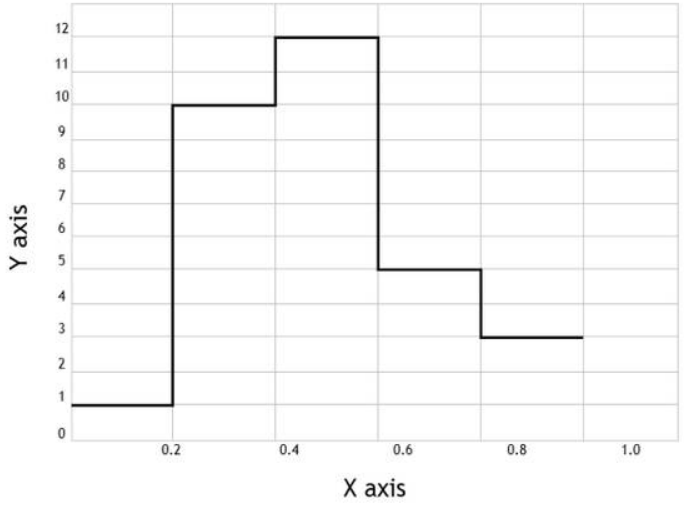
Type MS · The internet, the web, DNS and the parts of a URL

Answer 1 of 7 · Question 3 part b.iv · 4 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part b.iv in this mark scheme extract

<table>
<tr><td>3(a)(i)</td><td>Award one mark for each of:

• At least four conversions plotted at the correct amplitude (1)
• Correct start point (1) 0 – 1 digital. 0 – 0 analogue
• Digital sound wave drawn (1) using candidate's plots

<table><tr><th>Sample number</th><th>Denary value</th></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>10</td></tr><tr><td>3</td><td>12</td></tr><tr><td>4</td><td>5</td></tr><tr><td>5</td><td>3</td></tr></table>
 
</td><td>MP1 does not have to start at 0.2 - 1</td><td>3</td></tr>
<tr><td>3(a)(ii)</td><td>Award one mark for:

• Time (1)
• Sample interval/period (1)
• Seconds (1)</td><td></td><td>1</td></tr>
<tr><td>3(a)(iii)</td><td>Award one mark for:

• Amplitude / sound level / volume (1)
• Metres/centimetres/nanometres (1)
• m/cm/nm (1)</td><td></td><td>1</td></tr>
</table>

Type MS · The internet, the web, DNS and the parts of a URL

Answer 1 of 7 · Question 3 part b.iv · 4 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part b.iv in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(i)</td><td>Award one mark from:
Transfer time would be longer (1)
Larger file size / less compression (1)
Takes up more storage space (1)
Uses more of her data allocation (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(ii)</td><td>Award up to two marks for a linked explanation such as:
Storage capacity can be scaled up and down (1) so no need to buy more secondary storage / only pay for what is used/needed (1)
Will have the files all in one place (1) rather than scattered across many secondary storage devices (1)
Files can be uploaded/downloaded anytime/anywhere/on any device (1) so long as there is an internet connection (1)
Can be set up to automatically backup / synchronise with mobile devices (1) therefore if any files/devices are lost/stolen her files will be available on the server (1)</td><td>Do not award a mark for cheaper without expansion.</td><td>2</td></tr>
<tr><td>3(b)(iii)</td><td>Award one mark from:
Storage host could be targeted by hackers (1)
Alyssa has less control over her files (1)
An untrustworthy employee (of the cloud storage provider) could steal her files (1)
Reliant on the storage provider for security / safekeeping (1)
Files could be intercepted/corrupted during upload/download (1)</td><td></td><td>1</td></tr>
<tr><td>3(b)(iv)</td><td>Award one mark for each of:<table><tr><th>URL component</th><th>Description</th></tr><tr><td>https</td><td>Protocol / scheme (1)</td></tr><tr><td>www.cloudisfab.com</td><td>Domain (name) / host / name of website (1)</td></tr><tr><td>re12</td><td>Folder/directory (on the website) / path / part of path(1)</td></tr></table></td><td></td><td>4</td></tr>
</table>

Type MS · The internet, the web, DNS and the parts of a URL

Answer 1 of 7 · Question 3 part b.iv · 4 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part b.iv in this mark scheme extract

	ru2.mp3	File/media/resource wanted (on the website)/ path / part of path (1)		
			Total for question 3	**13**

Type MS · The internet, the web, DNS and the parts of a URL

Answer 1 of 7 · Question 3 part b.iv · 4 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part b.iv in this mark scheme extract

Question Number	Answer (flow chart replaced)	Additional Guidance	Mark

Type MS · The internet, the web, DNS and the parts of a URL

Answer 2 of 7 · Question 2 part c · 2 marks

2019 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

2(a)	Award **one** from each: Zafer: 1. Choose strong (1) authentication credentials (username/password) 2. Not sharing his credentials (1) 3. Not having automatic login to cloud / leaving his machine unattended (1) 4. Changing password regularly (1) 5. Password protect documents (1) Provider: 1. Infrastructure (e.g. firewall/servers) must be secure from unauthorised access (1) 2. Keeping their security software up to date (1) 3. Policy and procedures effective in preventing an insider attack / a data breach (1) 4. Data protection laws in the resident country must be obeyed (1) 5. Backup and restore procedures (1) 6. Encryption (1)		2

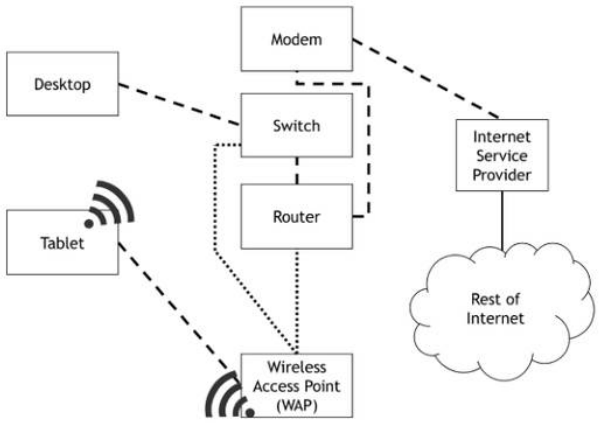
Question Number	Answer	Additional Guidance	Mark
2(b)	**Two** marks if completely correct. **One** mark for any two correct cells. **Robert** / **Zafer** Washing machine design drawings: Read / Read A folder of manuals for new machines: None / Read, Write Or Write, Read	Ignore spelling mistakes	2

Type MS · The internet, the web, DNS and the parts of a URL

Answer 2 of 7 · Question 2 part c · 2 marks

2019 JUN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(c)	Award **one** mark from each category (maximum of 2) Input: • www.pearson.com / website domain (1) • Uniform Resource Locator / URL (of Pearson website) (1) Output: • 2.20.38.113 (1) • Pearson's machine IP(v4) address (1)	Don't penalise spelling or typos, especially in addresses	2

Question Number	Answer	Additional Guidance	Mark
2(d)	One mark for each dotted connection, correctly located. Maximum of 6 marks. Note: WAP can be connected to either the Switch or the Router, but not both  		6

Type MS · The internet, the web, DNS and the parts of a URL

Answer 2 of 7 · Question 2 part c · 2 marks

2019 JUN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The internet, the web, DNS and the parts of a URL

Answer 3 of 7 · Question 1 part c · 2 marks

2020 NOV · Paper 01 · extract 1 of 5 · find part c in this mark scheme extract

1(a)	Any **two** from: • (Share) access to the Internet/WWW/broadband connection (1) • (Internal) communication (using email/instant messaging/calendars) (1) • Share files/data (1) • Share peripherals/printers / hardware (1) • Increases the amount of storage (since network storage devices can be attached) (1) • Saves money on licences (since network site licences for software are usually cheaper than buying a stand-alone licence for every machine) (1) • Centralised backup (1) • Centralised security (1)	• Award examples if mapped to a bullet point • Do not award the same bullet more than once • Do not award for install updates (in the stem)	2

Question Number	Answer	Additional Guidance	Mark
1(b) (i)	Any **two** from: • ~~Faster (data transfer) (1)~~ Not really true, its Greater bandwidth / more bits per second (1) • The connection does not get worse the further you are from the router / more reliable (1) • Connection does not get obstructed by walls, ceilings, and furniture (1) • More secure (1)	Accept reverse arguments for disadvantages of wireless.	2

Type MS · The internet, the web, DNS and the parts of a URL

Answer 3 of 7 · Question 1 part c · 2 marks

2020 NOV · Paper 01 · extract 2 of 5 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b) (ii)	Any **two** from: • Installation/maintenance is more complex (1) • Devices need to be physically connected (1) • Less portable / limited by length of cable (1) • Limited number of devices can be connected (1) • Some digital devices cannot use a wired connection (1) • Trip hazard (1)	Accept reverse arguments for advantages of wireless.	2

Question Number	Answer	Additional Guidance	Mark
1(c)	Award **two** marks for a linked description that addresses the individual mark points: The internet is a (global) network of networks/ connected devices (1) whereas the www is the collection of web pages/service accessed using the internet (1) / the www is resources located via URLs/domain names (1) The internet is the infrastructure (1) and the www is a service that runs on that infrastructure (1)		2

Question Number	Answer	Additional Guidance	Mark

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Type MS · The internet, the web, DNS and the parts of a URL

Answer 3 of 7 · Question 1 part c · 2 marks

2020 NOV · Paper 01 · extract 3 of 5 · find part c in this mark scheme extract

1(d)	Award **two** marks for a linked explanation such as: Running out of IPv4 addresses/the number of (internet) connected devices has grown (1) IPv6 addresses are long/longer than IPv4 / the number of possible addresses is large/will last much longer (1)		2

Question Number	Answer	Additional Guidance	Mark
1(e)(i)	Star		1

Question Number	Answer	Additional Guidance	Mark
1(e) (ii)	Any **one** from: • easy to connect/remove new nodes (1) • failure of one node/link does not affect the rest of the network (1) • easy to detect the failure of one node/link (1)		1

Question Number	Answer	Additional Guidance	Mark
1(e) (iii)	Any **one** from: • if central switch/hub fails then the whole network fails (1) • performance and number of nodes that can be added depends on capacity of central switch/hub (1) • can require more cable than some of the other topologies (1)		1

Question Number	Answer	Additional Guidance	Mark

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Type MS · The internet, the web, DNS and the parts of a URL

Answer 3 of 7 · Question 1 part c · 2 marks

2020 NOV · Paper 01 · extract 4 of 5 · find part c in this mark scheme extract

1(f)(i)	Any **two** from: • SMTP (1) • IMAP (1) • POP/POP3 (1)		2

Question Number	Answer	Additional Guidance	Mark
1(f)(ii)	Any **two** from: • passes the (reassembled) packets to the application layer (1) • check if all packets have arrived (1) • determine whether the contents are correct (1) • requests resending of lost or damaged packets (1) • reassembles packets in correct order/ into an email (1)	Do not accept receives from network layer (stem)	2

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · The internet, the web, DNS and the parts of a URL

Answer 3 of 7 · Question 1 part c · 2 marks

2020 NOV · Paper 01 · extract 5 of 5 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The internet, the web, DNS and the parts of a URL

Answer 4 of 7 · Question 2 part d · 3 marks

2021 JUN · Paper 01 · extract 1 of 5 · find part d in this mark scheme extract

Number			
2(a)	Award **one** mark from: • share peripherals/devices (e.g. printers, scanners) (1) • communicate (e.g. email, instant messaging, play games) (1) • share data (e.g. files/music/videos/backups on servers) (1) • deploy/update applications/software(1) • administrative purposes (e.g. remote access/desktop)		1

Question Number	Answer	Additional Guidance	Mark
2(b)	Award **one** mark from: • to allow (connected) machines to communicate (1) • to provide the rules of communication between (networked) devices (1)	Accept any other response indicating protocols enable communication between devices	1

Type MS · The internet, the web, DNS and the parts of a URL

Answer 4 of 7 · Question 2 part d · 3 marks

2021 JUN · Paper 01 · extract 2 of 5 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	**The only correct answer is A** ***B*** *is not correct because FTP is an application protocol, which is added in the application layer* ***C*** *is not correct because FTP is an application protocol, which is added in the application layer* ***D*** *is not correct because FTP is an application protocol, which is added in the application layer*		1
2(c)(ii)	Award **one** mark from: • Adds the source/destination IP address (1) • Establishes sockets (an IP address followed by a port number) (1) • Moves packets onto the next network node (1) • Strips the source/destination IP addresses (when it arrives at its destination) (1) • Passes the packet to the transport/next layer (when it arrives at its destination) (1) • Adds/removes packet headers (1)		1

Question Number	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for each of: **Domain Name** / **IP Address** / **URL** (blank) / (blank) / ✓ (blank) / ✓ / (blank) ✓ / (blank) / (blank)		3

Type MS · The internet, the web, DNS and the parts of a URL

Answer 4 of 7 · Question 2 part d · 3 marks

2021 JUN · Paper 01 · extract 3 of 5 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(e)(i)	Bus		1
2(e)(ii)	Award **one** mark from: • Easy to connect/remove devices/computers (in a network) (1) • Cable cost less / fewer cables needed (compared to mesh/star topology) (1) • Easier to plan/setup/maintain (1)		1
2(e)(iii)	Award **one** mark from: • If the main cable/a terminator fails or gets damaged the whole network will fail (1) • The more workstations the slower it gets (1) • The more workstations the more collisions (1) • Every device 'sees' all of the data on the network (1)		1

Question Number	Answer	Additional Guidance	Mark
2(f)(i)	Award **two** marks from: • Reduces time taken to upload/download (1) • Reduces the bandwidth required/cost of data to upload/download (1) • (Usually) produces smaller file/reduces storage (on the web server/device) (1) • People have a limited hearing range / can't tell the difference (1) • People's brains will 'fill in the blanks' (1)		2
2(f)(ii)	**The only correct answer is C** ***A*** *is not correct because it is not compressed* ***B*** *is not correct because it uses lossless compression* ***D*** *is not correct because it uses lossless compression*		1
2(f)(iii)	Award **one** mark from: • 3c3w2c6w3c (1) • c3w3c2w6c3 (1)		1

Type MS · The internet, the web, DNS and the parts of a URL

Answer 4 of 7 · Question 2 part d · 3 marks

2021 JUN · Paper 01 · extract 4 of 5 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
2(g)	Award **four** marks for a correct expression or correct answer **Examples of correct expressions:** 6 x 1000 x 1000 x 1000 x 8 (4) 54 x 1000 x 1000 1000 x 8 (4) 9 888.9 (4) For an incorrect or partial expression award **one** mark for each of: • **Bits to transfer** ○ 6×1000^3 (1) ○ x 8 (or equivalent) (1) • **Speed in bits per second** ○ 54×1000^2 (1) • Their bits divided by their speed (1)	• Units are not required • Equivalent expressions are accepted • Allow follow through for incorrect or partial expressions	4

Type MS · The internet, the web, DNS and the parts of a URL

Answer 4 of 7 · Question 2 part d · 3 marks

2021 JUN · Paper 01 · extract 5 of 5 · find part d in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The internet, the web, DNS and the parts of a URL

Answer 5 of 7 · Question 1 part e.i · 1 mark

2024 JUN · Paper 01 · extract 1 of 3 · find part e.i in this mark scheme extract

Number			
1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: • WANs cover large/wide (geographic) distances/areas / LANs cover small/local (geographic) distances/areas (1) • WANs have lower bandwidth / LANs have higher bandwidth (1) • WANs connect multiple LANs / LANs are single networks (1) • WANs have higher latency / LANs have lower latency (1) • WANs use shared infrastructure / LANs are usually privately owned (1) • WANs use third party connections / LANs are owned by the organisation (1) • WANs have more users/devices / LANs have fewer users (1)	Do not award 2 marks for the same point e.g. A WAN is used over a large geographic area whereas a LAN covers a small distance. Two different differences must be identified. Do not accept general points such as WANs are slower - too vague. Do not accept LANs can be wireless.	2
1(b)	Award **one** mark for any of the following: • (Increased risk of) malware/viruses **spreading** / Pharming / Cyber attack (1) • (Increased risk of) hacking/unauthorised access/interception (1)		1
1(c)(i)	Award **one** mark for: • (Partially connected) mesh (1)		1

Type MS · The internet, the web, DNS and the parts of a URL

Answer 5 of 7 · Question 1 part e.i · 1 mark

2024 JUN · Paper 01 · extract 2 of 3 · find part e.i in this mark scheme extract

1c(ii)	Award **one** mark for any of the following: • More expensive to install (1) • Needs more cabling/wiring (1) • Requires more maintenance (1) • Harder to maintain / manage / install (1) • Higher power consumption (1)		1
1(c)(iii)	Award up to **two** marks for a linked explanation such as: • In the event of a damaged/congested connection data/packets can still be routed to their destination (1) because each node/computer is connected to at least two other nodes / there are multiple pathways available (1)		2
1(d)	**The only correct answer is D** *A is not correct because the application layer deals with applications* *B is not correct because the transport layer deals with packet structure* *C is not correct because the network layer deals with routing end to end*		1
1(e)(i)	Award **one** mark for: • https/hypertext transfer protocol secure (1)		1
1(e)(ii)	Award **one** mark for: • specification(.html) (1)		1
		Total for question 1	**10**

Type MS · The internet, the web, DNS and the parts of a URL

Answer 5 of 7 · Question 1 part e.i · 1 mark

2024 JUN · Paper 01 · extract 3 of 3 · find part e.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The internet, the web, DNS and the parts of a URL

Answer 6 of 7 · Question 1 part e.ii · 1 mark

2024 JUN · Paper 01 · extract 1 of 3 · find part e.ii in this mark scheme extract

Number			
1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: • WANs cover large/wide (geographic) distances/areas / LANs cover small/local (geographic) distances/areas (1) • WANs have lower bandwidth / LANs have higher bandwidth (1) • WANs connect multiple LANs / LANs are single networks (1) • WANs have higher latency / LANs have lower latency (1) • WANs use shared infrastructure / LANs are usually privately owned (1) • WANs use third party connections / LANs are owned by the organisation (1) • WANs have more users/devices / LANs have fewer users (1)	Do not award 2 marks for the same point e.g. A WAN is used over a large geographic area whereas a LAN covers a small distance. Two different differences must be identified. Do not accept general points such as WANs are slower - too vague. Do not accept LANs can be wireless.	2
1(b)	Award **one** mark for any of the following: • (Increased risk of) malware/viruses **spreading** / Pharming / Cyber attack (1) • (Increased risk of) hacking/unauthorised access/interception (1)		1
1(c)(i)	Award **one** mark for: • (Partially connected) mesh (1)		1

Type MS · The internet, the web, DNS and the parts of a URL

Answer 6 of 7 · Question 1 part e.ii · 1 mark

2024 JUN · Paper 01 · extract 2 of 3 · find part e.ii in this mark scheme extract

1c(ii)	Award **one** mark for any of the following: • More expensive to install (1) • Needs more cabling/wiring (1) • Requires more maintenance (1) • Harder to maintain / manage / install (1) • Higher power consumption (1)		1
1(c)(iii)	Award up to **two** marks for a linked explanation such as: • In the event of a damaged/congested connection data/packets can still be routed to their destination (1) because each node/computer is connected to at least two other nodes / there are multiple pathways available (1)		2
1(d)	**The only correct answer is D** *A is not correct because the application layer deals with applications* *B is not correct because the transport layer deals with packet structure* *C is not correct because the network layer deals with routing end to end*		1
1(e)(i)	Award **one** mark for: • https/hypertext transfer protocol secure (1)		1
1(e)(ii)	Award **one** mark for: • specification(.html) (1)		1
		Total for question 1	**10**

Type MS · The internet, the web, DNS and the parts of a URL

Answer 6 of 7 · Question 1 part e.ii · 1 mark

2024 JUN · Paper 01 · extract 3 of 3 · find part e.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · The internet, the web, DNS and the parts of a URL

Answer 7 of 7 · Question 1 part a · 4 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part a in this mark scheme extract

<table>
<tr><td>1(a)</td><td>Award one mark for each correct row:

<table><tr><th>Statement</th><th>DNS</th><th>PAN</th><th>WAN</th><th>WWW</th></tr><tr><td>A network of networks not limited to a single location</td><td></td><td></td><td>✓</td><td></td></tr><tr><td>A network that connects devices within the range of a single user</td><td></td><td>✓</td><td></td><td></td></tr><tr><td>A set of hypertext documents accessed through the HTTP protocol</td><td></td><td></td><td></td><td>✓</td></tr><tr><td>A system that maps domain names to IP addresses</td><td>✓</td><td></td><td></td><td></td></tr></table></td><td>Do not award a mark for a row containing more than one tick.</td><td>4</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(b)</td><td>Award one mark for each of:

• Transport (layer) (1)
• (Data) link / network link / network interface (layer) (1)</td><td>Accept
• transfer or transmission for Transport

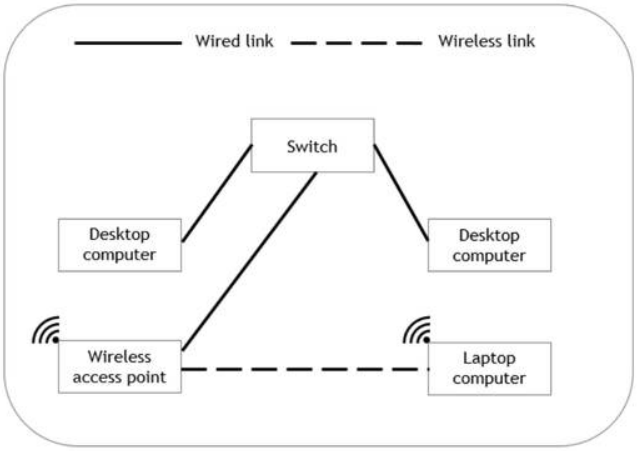
Do not accept:
• Network layer and Application layer (in the question)
• Data access layer, Internet layer (alternative names for network layer)
• Data’ or ‘network’ without ‘link’ or ‘interface’</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(c)</td><td>The only correct answer is D

A is not correct because HTTP is used to transfer hypertext
B is not correct because it is used to view email held on a server
C is not correct because it is used to download email from a server</td><td></td><td>1</td></tr>
</table>

Type MS · The internet, the web, DNS and the parts of a URL

Answer 7 of 7 · Question 1 part a · 4 marks

2025 JUN · Paper 01 · extract 2 of 4 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each correct connection: • One solid line from wireless access point to switch (1) • One solid line from each desktop to switch (1) • One dashed line from laptop computer to wireless access point (1) 	No marks should be awarded if any additional lines have been drawn for each mark point. Do not double penalise (e.g. if you have penalised a second line from desktop to WAP then don't penalise extra line from WAP to desktop)	3

Type MS · The internet, the web, DNS and the parts of a URL

Answer 7 of 7 · Question 1 part a · 4 marks

2025 JUN · Paper 01 · extract 3 of 4 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark
1(e)	Award up to **two** marks for a linked description such as: • To find weaknesses in a system (1) fix/repair weakness (1) • To find vulnerabilities in a system (1) by testing the effectiveness of the security measures in place (1) • An ethical hacking attack on a network (1) to identify vulnerabilities / recommend measures/take action to improve security (1) • Tests whether users of the network (1) can be tricked into giving away information / giving criminals access to the network / downloading malware (1) • To test the system security (1) / sanctioned by the owner (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
		Total for question 1	**12**

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Type MS · The internet, the web, DNS and the parts of a URL

Answer 7 of 7 · Question 1 part a · 4 marks

2025 JUN · Paper 01 · extract 4 of 4 · find part a in this mark scheme extract

Question	Answer	Additional Guidance	Mark